Salahaddin University-Erbil
College of Computer Science and IT
Stage 2 Cs & IT

Web Technology Basics - Lec. 4 – HTML5

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Tuesday: IT- 8:30-10:30/ Hall 3
CS- 10:30 -12:30/ Hall 16

What you will gain from this course:

- Understand how websites work (browser ↔ server).
- Build simple, semantic pages with HTML5.
- Style and layout pages with CSS.
- Make pages responsive, accessible, and performant.

HTML5

- HTML5 is the latest version of HTML that comes with syntax compatible with HTML4 and XHTML.
- HTML5 has many flexible features.
- HTML5 syntax is easy.

•For tag attributes, quoting is not mandatory. Also, attribute values are not obligatory.
•It is also not mandatory to close any empty void elements.
•Only void elements (BR, IMG, LINK, etc.) can be self-closed using this />.
•Upper-case tag names are also allowed.
•The starting tags are not mandatory for every element. (not good practice)
Note: In HTML5, the start tag of certain elements is optional because the browser can infer it from the document structure, but this applies only to specific elements and not to all HTML tags.
<html>, <head>, <body>

HTML5

Block vs Inline Elements

- An HTML document consists of HTML block-level or inline elements.
- An HTML block-level element always
  - starts on a new line and spans left and right, As far as possible.
  - It occupies the entire horizontal space of the parent element, whose height is equal to the HTML content.
  - Example of block level elements:<address>, <article>,<aside>, <blockquote>, <canvas>, <dd>, , <dl>, <dt>, <fieldset>, <figcaption>, <figure>, <footer>, <form>, <h1> to <h6>, <header>,<hr>, <li>, <main>, <nav>, <noscript>, <ol>, <p>, <pre>, <section>, <table>, <tfoot>, <ul>, and <video>
- An inline element
  - Does not start on a new line.
  - An inline element only takes up as much width as necessary.
  - <a><abbr><acronym><b><bdo><big>
<button><cite><code><dfn><i><img><input><kbd><label><map><object><output><q><samp><script><select><small><sub><sup> <textarea><time><tt><var>

EXAMPLE:

```
<p>This is an inline span <span style="border: 1px solid black">Hello World</span> element inside a paragraph.</p>
```


HTML5

HTML Comments

- HTML comments are not displayed in the browser, but they can help document your HTML source code.

```
<!-- Write your comments here -->
```

HTML Symbols

- Symbols or letters that are not present on your keyboard can be added to HTML using entities.
- To add such symbols to an HTML page, you can use the entity name or the entity number (a decimal or a hexadecimal reference) for the symbol.

EXAMPLE:

```
<p>I will display &euro;</p>
<p>I will display &#8364;</p>
<p>I will display &#x20AC;</p>
```

I will display €
I will display €
I will display €

HTML Emjoi

- Emojis look like images, but they are not.
- Emojis are letters (characters) from the UTF-8 (Unicode) character set:

EXAMPLE:

```
<p> &#128513;</p>
<p>I will display &#9996;</p>
```

😁
I will display ✌️

HTML5

<code> tag

- The <code> tag is used to insert variables, fragments of program code, etc. into an HTML document.
- In the browser, the code is displayed in a monospaced font (a font in which all characters have the same width) of the smaller size.
- The <code> tag alone represents a single code line or code phrase. It should be wrapped within a <pre> element for representing several lines of code.

```
<p>Example1 of Code tag </p>
<p>To create a hyperlink, you can use the following HTML code:</p>
<code> &lt;a href="//www.w3schools.in/"&gt;Link Text&lt;/a&gt; </code>
<p>Example2 of Code tag </p>
<pre>      <code>
                body {
                  color: yellow;
                  font-size: 16px;
                  line-height: 1.5;
                }
            </code>
</pre>
```

Example1 of Code tag

To create a hyperlink, you can use the following HTML code:

```
<a href="//www.w3schools.in/">Link Text</a>
```

Example2 of Code tag

```
body {
  color: yellow;
  font-size: 16px;
  line-height: 1.5;
}
```


HTML5

<embed> tag

- The <embed> tag in HTML is used to embed external resources, such as videos, games, or audio files, into an HTML document. It is an empty tag.

```
<html>
 <head> <title>My Video</title> </head>
 <body>
  <h1>My Video</h1>
  <embed src="myvideo.mp4" type="video/mp4"
width="640" height="360" autoplay>
 </body>
</html>
```

```
<!-- Embed an audio file -->
 <embed src="audiofile.mp3" type="audio/mpeg">

<!-- Embed a Flash game -->
 <embed src="game.swf" type="application/x-
shockwave-flash">

<!-- Embed a PDF -->
 <embed src="document.pdf"
type="application/pdf" width="800" height="600" />
```

HTML5

<picture> tag

- The HTML <picture> tag is an essential element for responsive web design. It lets developers specify multiple source files for an image, allowing the browser to choose the most appropriate file based on the current viewport, screen resolution, and other factors.
- The <picture> tag works in conjunction with the <source> and <img> tags. Here's how these tags work together:
- The <picture> tag is a container that wraps the structure. It tells the browser that an image with multiple source files is available.
- The <source> tag is used to specify different image files. Each <source> tag includes a media attribute specifying the conditions for using that source file.
- The <img> tag is a default image. It is used if none of the <source> tags match the current conditions.
- The question may be, "Why not just use the <img> tag?" The <img> tag is excellent with a single image source. But concerning responsive design, the <picture> tag provides more flexibility by allowing multiple sources to be specified.

```
<picture>
 <source media="(min-width:650px)" srcset="Kurdistan flag.png">
 <source media="(min-width:465px)" srcset="kurdistan-emoji.webp">
 <img src="big flag.jpg" alt="Flowers" style="width:auto;">
</picture>
```


HTML5

< canvas > tag

The HTML <canvas> tag is a rectangular space with specific height and width attributes dedicated to graphical content that allows you to create graphics in a web browser. It provides multiple ways to draw shapes, text, lines, circles, and images.
Canvas is widely used for game development, photo editing, graphic designing, animation, data visualization, and video processing. However, the canvas initially has no border or content, and you need to use JavaScript to create them.

```
<canvas id="myCanvas" width="300" height="150" style="border:1px solid grey"></canvas>
    <script>
    const myCanvas = document.getElementById("myCanvas");
    const ctx = myCanvas.getContext("2d");
    ctx.fillStyle = "red";
    ctx.fillRect(20, 20, 150, 100);
    </script>
```

HTML5

< SVG > HTML Graphics

- Scalable Vector Graphics (SVG) is an XML-based language for creating vector graphics. It's an essential part of modern web development.
- SVG graphics are scalable, and do not lose any quality if they are zoomed or resized, and it is supported by all major browsers.

```
<svg width="300" height="130" style="border:2px solid black">
  <rect width="200" height="100" x="10" y="10" rx="20" ry="20" fill="green" stroke="red"
        stroke-width="4"/>
  <text x="110" y="70" fill="none" stroke="yellow" rotate="-20" font-size="30px"  text-
        anchor="middle" >KURDISTAN</text>
</svg>
```

- **x** and **y** attributes defines the x- and y-position of the top-left corner of the rectangle (x="10" places the rectangle 10px from the left margin of the SVG border and y="10" places the rectangle 10px from the top margin) in the SVG canvas.
- **rx** and **ry** attributes defines the radius of the corners of the rectangle.
- **fill** attribute defines the fill color of the rectangle.
- **stroke-width** property defines the width of the border of the rectangle
- **stroke** property defines the color of the border of the rectangle

HTML5-SVG Filters

- SVG filters are used to add special effects to SVG graphics.
- All SVG filters are defined within a <defs> element. The <defs> element is short for "definitions", and contains definitions of special elements (such as filters).
- The <filter> element is used to define an SVG filter. It has a required id attribute which identifies the filter.
- Inside the <filter> element, we put one or more filter effects to use on the graphic.

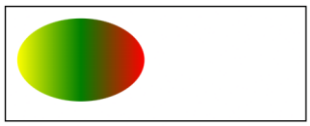
```
<svg height="110" width="110">
 <defs>
  <filter id="f1">
   <feGaussianBlur in="SourceGraphic" stdDeviation="0"/>
   <feDropShadow dx="9" dy="9" stdDeviation="3" flood-opacity="1"
         flood-color="black"/>
  </filter>
 </defs>
<circle r="45" cx="50" cy="50" fill="yellow" stroke="green"
         filter="url(#f1)">
</svg>
```

- **blur** effect is defined with the <feGaussianBlur> element
- **in="SourceGraphic"** part defines that the effect is created for the entire element
- **stdDeviation** attribute defines the amount of the blur
- **filter** attribute of the <rect> element points the element to the "f1" filter.
- **dx** attribute defines the x offset of the drop shadow
- **dy** attribute defines the y offset of the drop shadow
- **flood-opacity** attribute defines the opacity of the drop shadow (from 0 to 1)

HTML5- SVG Gradients

- A gradient is a smooth transition from one color to another. In addition, several color transitions can be applied to the same element. There are two types of gradients in SVG:
  - Linear gradients - defined with <linearGradient>
  - Radial gradients - defined with <radialGradient>

```
<svg height="150" width="400">
  <defs>
    <linearGradient id="grad2" x1="0%" x2="100%" y1="0%" y2="0%">
      <stop offset="0%" stop-color="yellow" />
      <stop offset="50%" stop-color="green" />
      <stop offset="100%" stop-color="red" />
    </linearGradient>
  </defs>
  <ellipse cx="100" cy="70" rx="85" ry="55" fill="url(#grad2)" />
</svg>
```

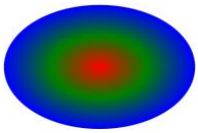


- **x1, x2, y1, y2** attributes of the <linearGradient> element define the x and y starting and ending points of the gradient
- **<stop>:** The colors of a gradient are defined with two or more <stop> elements
- **top-color** attribute in <stop> defines the color of the gradient stop
- **offset** attribute in <stop> defines where the gradient stop is placed
- **fill** attribute of the <ellipse> element points the element to the "grad1" gradient

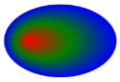
HTML5- SVG Gradients

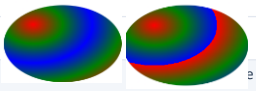
Radial gradient

```
<svg height="150" width="400" >
  <defs>
    <radialGradient id="grad3" cx="50%" cy="50%" r="50%" fx="50%" fy="50%">
      <stop offset="0%" stop-color="red" />
      <stop offset="50%" stop-color="green" />
      <stop offset="100%" stop-color="blue" />
    </radialGradient>
  </defs>
  <ellipse cx="100" cy="70" rx="85" ry="55" fill="url(#grad3)" />
</svg>
```



- **cx** The x position of the end circle of the radial gradient. Default is 50%
- **cy** The y position of the end circle of the radial gradient. Default is 50%
- **fx** The x position of the start circle of the radial gradient. Default is 50% →20%
- **fy** The y position of the start circle of the radial gradient. Default is 50%
- **r** The radius of the end circle of the radial gradient. Default is 50%
- **spreadMethod** Defines the spread method of the gradient. Possible values: "pad", "reflect", "repeat".



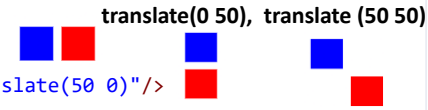


HTML5- SVG Transformation

- SVG elements can be manipulated using transform functions. The transform attribute can be used with any SVG element.
- The transform attribute defines a list of transform functions that can be applied to an element and the element's children: translate(), scale(), rotate(), skew()

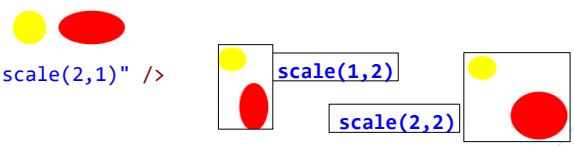
1. The **translate()** function is used to move an object by x and y.

```
<svg width="200" height="100" >
  <rect x="5" y="5" width="40" height="40" fill="blue" />
  <rect x="5" y="5" width="40" height="40" fill="red" transform="translate(50 0)"/>
</svg>
```



2. The **scale()** function is used to scale an object by x and y. If y is not provided, it is set to be equal to x. It is used to change the size of an object. It takes two numbers: the x scale factor and the y scale factor. For example, 0.5 shrinks the object by 50%.

```
<svg width="200" height="100" >
  <circle cx="25" cy="25" r="20" fill="yellow" />
  <circle cx="50" cy="25" r="20" fill="red" transform="scale(2,1)" />
</svg>
```



HTML5- SVG Transformation

3. The **rotate()** function is used to rotate an object by a degree.

```
<svg width="200" height="100" >
  <rect x="50" y="5" width="40" height="40" fill="blue" transform="rotate(45)" />
</svg>
```

4. The **skewX()** function is used to skew an object along the y axis by a degree.

```
<svg width="200" height="50" >
  <rect x="5" y="5" width="40" height="40" fill="blue" transform="skewX(30)" />
</svg>
```

- **skewY()** function is used to skew an object along the y axis by a degree.

```
<svg width="200" height="50">
  <rect x="5" y="5" width="40" height="40" fill="blue" transform="skewY(30)" />
</svg>
```